Quantum Computing for Science

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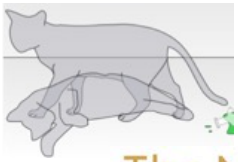
Email: anakano@usc.edu



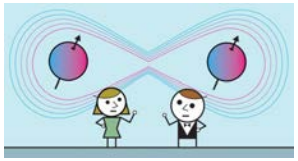
A fundamentally new way of computing — use it for science



What Is Quantum Computing?



Quantum computing utilizes quantum properties such as superposition & entanglement for computation



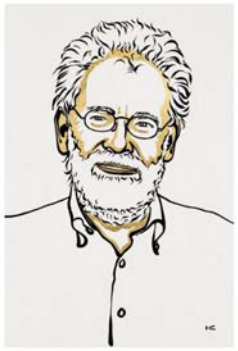
The Nobel Prize in Physics 2022



Ill. Niklas Elmehed © Nobel Prize Outreach
Alain Aspect
Prize share: 1/3



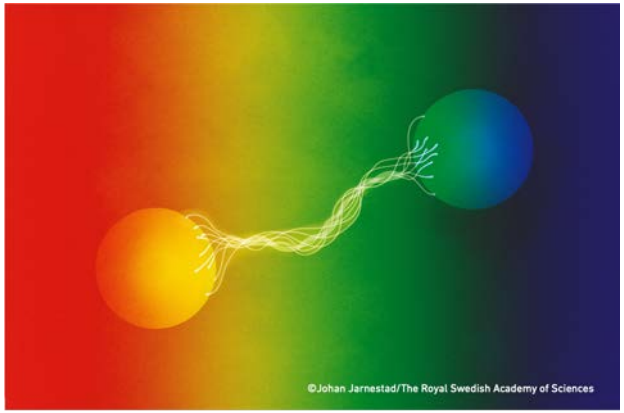
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John F. Clauser
Prize share: 1/3

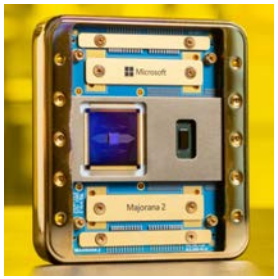


Ill. Niklas Elmehed © Nobel Prize Outreach
Anton Zeilinger
Prize share: 1/3

The Nobel Prize in Physics 2022 was awarded jointly to Alain Aspect, John F. Clauser and Anton Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science"

It's entanglement!





cf. Microsoft announced Majorana 1 quantum chip (Feb. 19, '25)

https://news.microsoft.com/azure-quantum

https://www.nsf.gov/science-matters/nsf-was-there-start-experimental-quantum-chip-may-yield-more

Quantum Computing Gets Hotter & Useful

Nobel Prize in Physics 2025



Ill. Niklas Elmehed © Nobel Prize Outreach
John Clarke
Prize share: 1/3



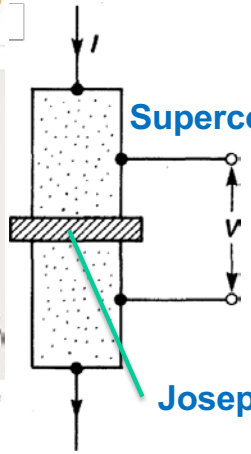
Ill. Niklas Elmehed © Nobel Prize Outreach
Michel H. Devoret
Prize share: 1/3



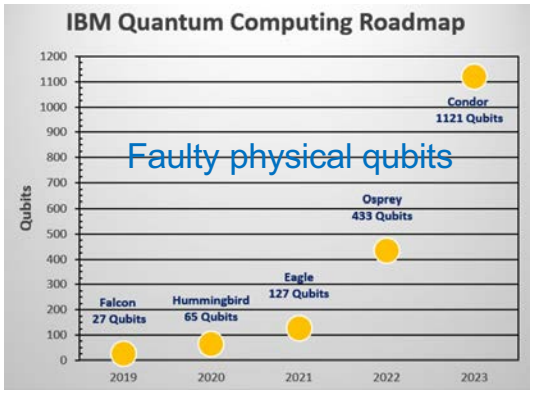
Ill. Niklas Elmehed © Nobel Prize Outreach
John M. Martinis
Prize share: 1/3

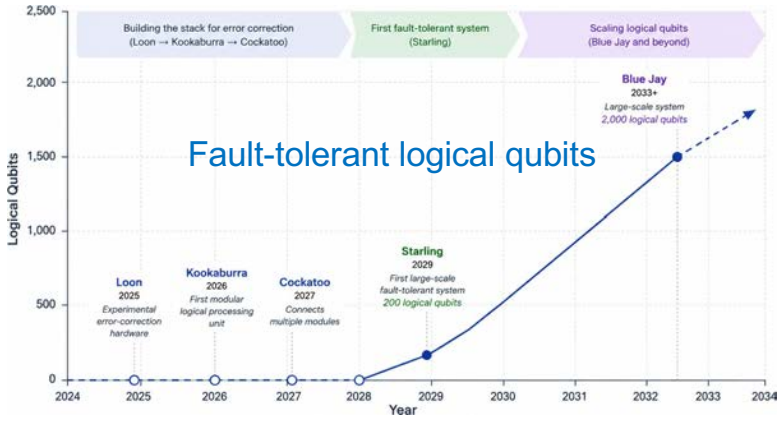
The Nobel Prize in Physics 2025 was awarded jointly to John Clarke, Michel H. Devoret and John M. Martinis "for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit"

Foundation of superconducting quantum circuits



Exponential growth of qubit counts

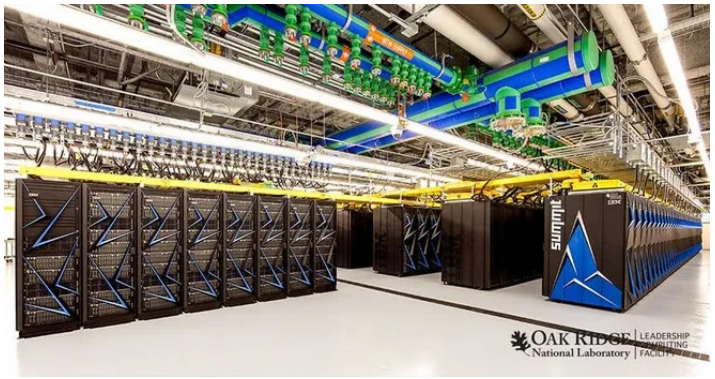




Quantum Computing Is Now

- **U.S. Congress (Dec. 21, ’18) signed National Quantum Initiative Act to ensure leadership in quantum computing & its applications**
- **Quantum advantage (*i.e.*, quantum computer is faster than the fastest supercomputer) was demonstrated by Google**
  F. Arute, J. M. Martinis *et al.*, *Nature* **574**, 505 (’19)
- **Google’s Sycamore quantum computer consumed 26 kilowatts of power to outperform the 13 megawatts Summit supercomputer at Oak Ridge National Lab, *i.e.*, 500 times more energy efficient**



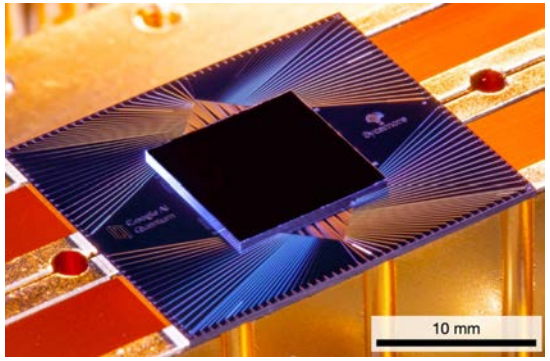


9,216-CPU & 27,648-GPU Summit

<

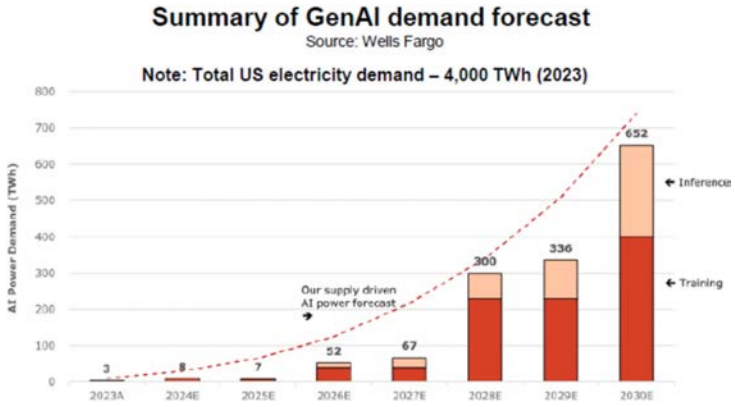
Fast & frugal

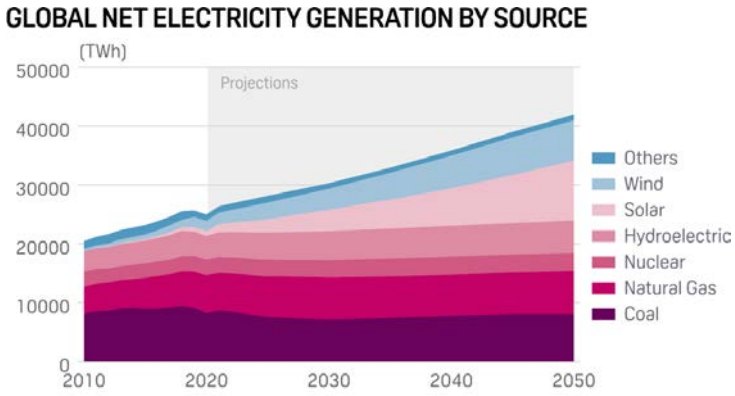




54-qubit Google Sycamore

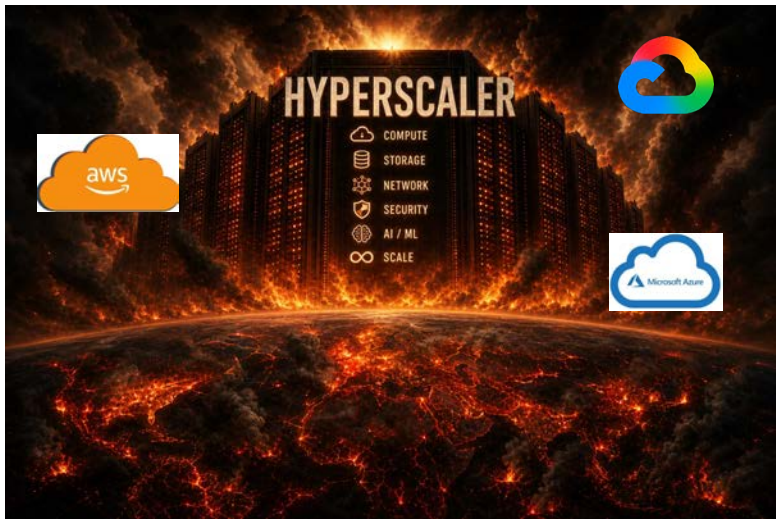
Why Care?





Source: EIA's International Energy Outlook 2021

- **Exponential growth of energy demand for AI-computing**



- **Linear growth of global energy supply (don't forget the growing CO_2 emission)**

- **Who will power your future AI-computing without burning out the Earth?**
- **Quantum computing may be the answer!**

Now AI & Quantum Computing for Science







Harness AI for breakthroughs in energy, science & national security

cf. United States and Japan Announce Historic $1 Billion Partnership (Jun. 4, ’26)







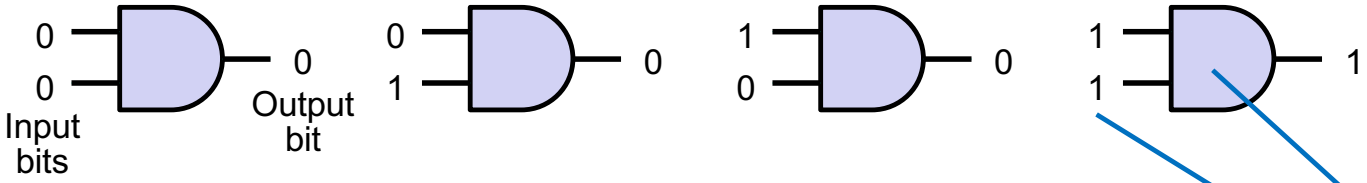
cf. National Quantum Initiative Act (Dec. 21, ’18)

What Is Computing?

- **Information is stored as a string of bits (0 or 1, *e.g.*, low or high voltage)**
- **Computer performs arithmetic (add, multiply, subtract, divide) & logical (and, or, not) operations on bit strings**
- **Example: logical AND gate**

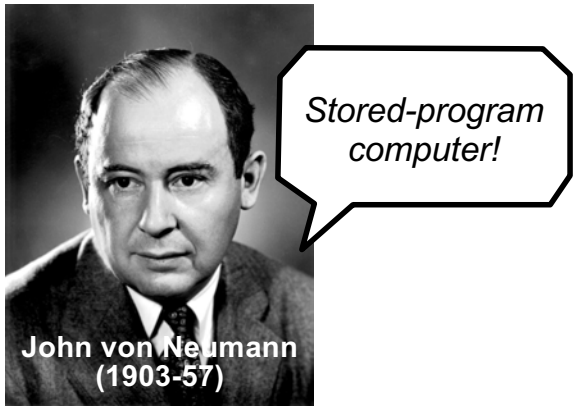


One of these input/output combinations happens at a time



von Neumann computer architecture (1944): logic gates & memory

data program





Processor

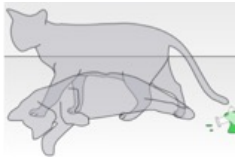


Memory

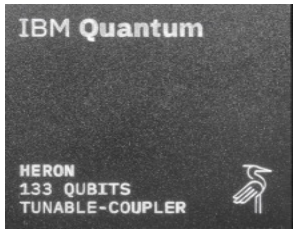
What Is Quantum Mechanics?

- **State of matter (*e.g.*, electron) is described by a complex-number wave function; its square is the probability to find the matter in a particular state**
- **Qubit: Superposition of 0 & 1 states with complex-number prefactors**

$$|\psi_{\text{qubit}}\rangle = a|0\rangle + b|1\rangle$$



$$|cat\rangle = a|\text{alive}\rangle + b|\text{dead}\rangle$$

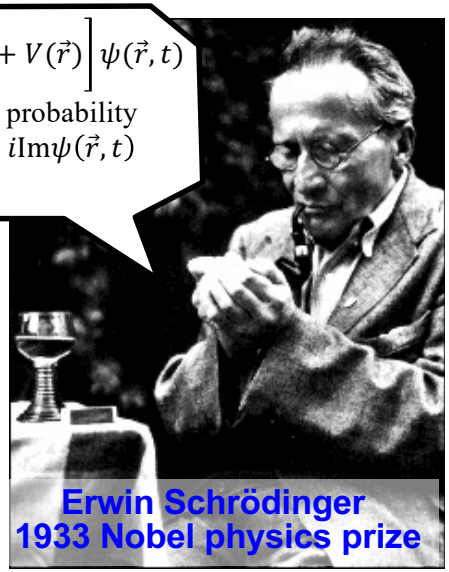


$$i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r})\right]\psi(\vec{r},t)$$

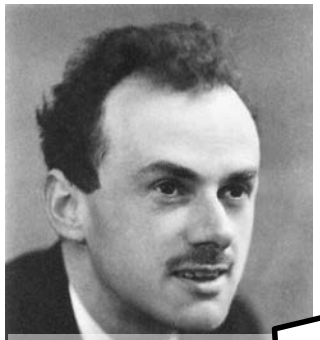
$$P(\vec{r},t) = |\psi(\vec{r},t)|^2 \sim \text{probability}$$

$$\psi(\vec{r},t) = \text{Re}\psi(\vec{r},t) + i\text{Im}\psi(\vec{r},t)$$

$$i = \sqrt{-1}$$



Erwin Schrödinger
1933 Nobel physics prize



Paul Dirac
1933 Nobel
physics prize

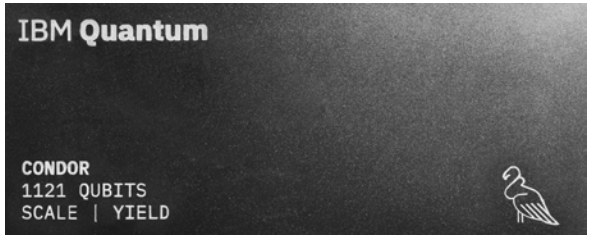
- **Quantum mechanics is exponentially hard — just describing *n*-qubit quantum state requires 2^n coefficients (*e.g.*, $2^{100}\sim10^{30}$)**

The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are completely known, and the difficulty is only that the exact application of these laws leads to equations ***much too complicated to be soluble****.*

What to do?

Now the Problem … and Solution

- Quantum mechanics is exponentially hard — just describing *n*-qubit quantum state requires 2^n coefficients (*e.g.*, $2^{100} \sim 10^{30}$)
- **But, the memory size of recent world's largest supercomputer, El Capitan at Lawrence Livermore National Lab, is merely 10^{14} double-precision words**





Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical.



Richard Feynman
1965 Nobel physics prize

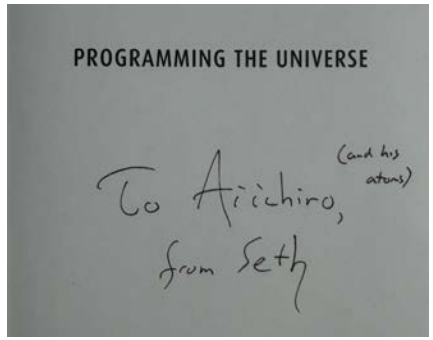
- **Feynman to the rescue: Let quantum mechanics compute itself, *i.e.*, quantum computing!** *Int. J. Theor. Phys.* **21**, 467 ('82)

Quantum Computational Science

- **An exciting scientific application of quantum computers is as a universal simulator of quantum many-body dynamics, as envisioned by Richard Feynman** *Int. J. Theor. Phys.* **21**, 467 ('82)
- **Seth Lloyd provided concrete algorithms and analysis** *Science* **273**, 1073 ('96)

  Watch Seth's movie: https://www.youtube.com/watch?v=EMzKshc6x2M

  https://www.amazon.com/Programming-Universe-Quantum-Computer-Scientist-ebook/dp/B000GCFBP6





- **Success in simulating *static* properties of quantum systems (*i.e.*, ground-state energy of small molecules)**

  Aspuru-Guzik *et al.*, *Science* **309**, 1704 ('05)
- **Challenge: Simulate quantum many-body *dynamics* on current-to-near-future noisy intermediate-scale quantum (NISQ) computers**

  J. Preskill, *Quantum* **2**, 79 ('18)
- **Second edition of *Feynman Lectures on Computation* (2023) added a section on "Simulating quantum dynamics" by John Preskill** *arXiv*:2106.10522 ('23)

https://www.amazon.com/Feynman-Lectures-Computation-Anniversary-Frontiers/dp/0367857332

Quantum Dynamics on Quantum Computers

- **We simulated nontrivial quantum dynamics on publicly available IBM's Q16 Melbourne & Rigetti's Aspen NISQ computers, *i.e.*, ultrafast control of emergent magnetism by THz radiation in 2D material** L. Bassman *et al.*, *Phys. Rev. B* **101**, 184305 ('20)





Lindsay
UChicago BS-CS
USC PhD-physics

https://www.planckian.co/



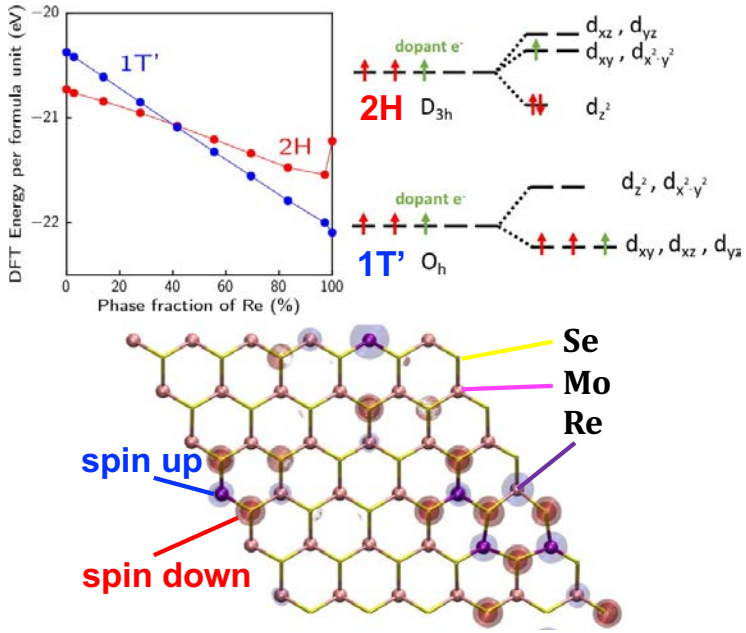
Miguel
USC sociology
→ BS-physics

Also, topological quantum dynamics

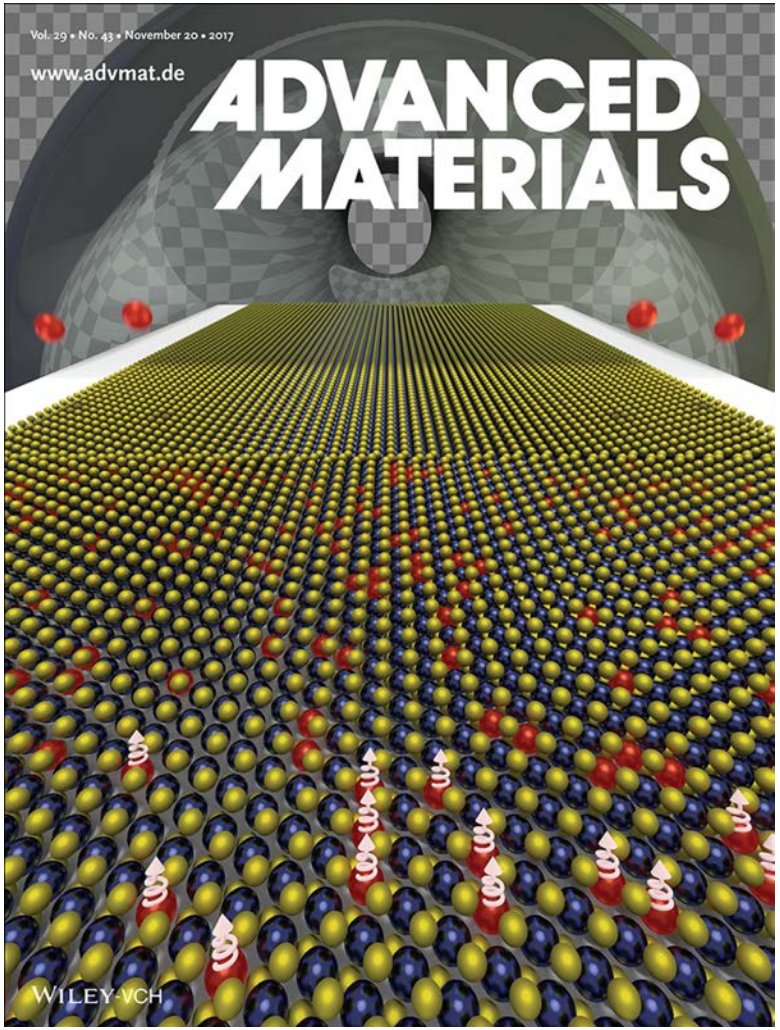
M. Mercado *et al.*, *Phys. Rev. B* **110**, 075116 ('24)
Y. Hang *et al.*, *Phys. Rev. B* **113**, 235145 ('26)

Application: Emergent Magnetism

- **Experiment at Rice shows 2H-to-1T' phase transformation by alloying $MoSe_2$ with Re**
- **QMD simulations at USC elucidate its electronic origin**
- **Simulation & experiment show novel magnetism centered at Re atoms**



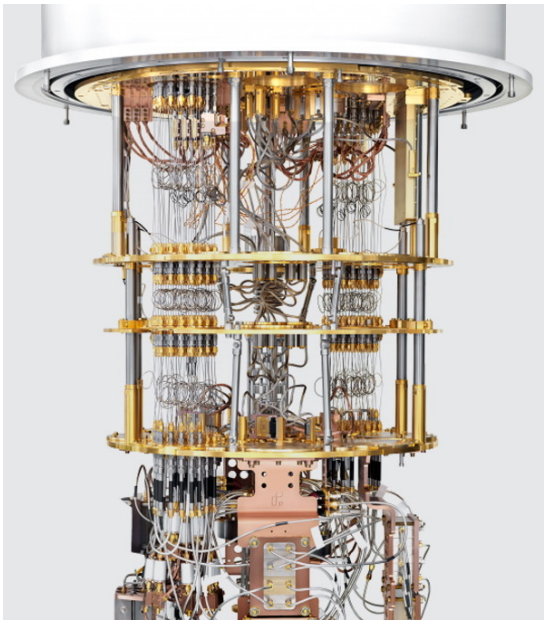
V. Kochat *et al.*, *Adv. Mater.* **29**, 1703754 ('17)



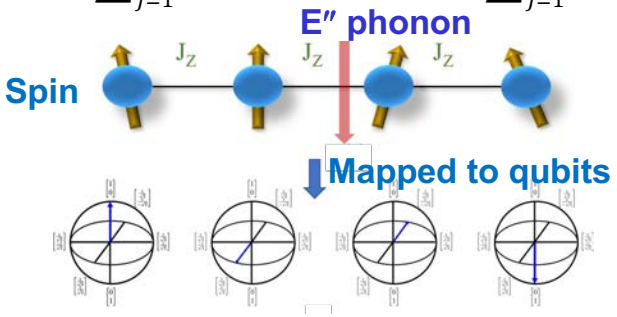
Quantum Computing of Magnetism

- **Simulated quantum many-body dynamics of magnetic spins on IBM's Q16 Melbourne & Rigetti's Aspen quantum processors**

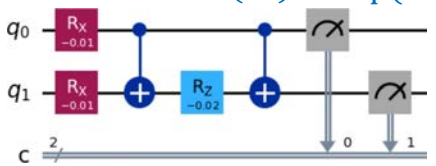
L. Bassman *et al.*, *Phys. Rev.* **101**, 184305 ('20)



$$H(t) = -J_z \sum_{j=1}^{N-1} \boldsymbol{\sigma}_z^j \boldsymbol{\sigma}_z^{j+1} - \varepsilon_{ph} \sin(\omega_{ph} t) \sum_{j=1}^{N} \boldsymbol{\sigma}_x^j$$



Quantum circuit: $U(\Delta t) = \exp(-iH\Delta t)$



Quantum (Python) program

```
circ.rx(-2*dt*B, 0)
circ.rx(-2*dt*B, 1)
circ.cx(0, 1)
circ.rz(-2*dt*J, 1)
circ.cx(0, 1)
```

Detailed explanation

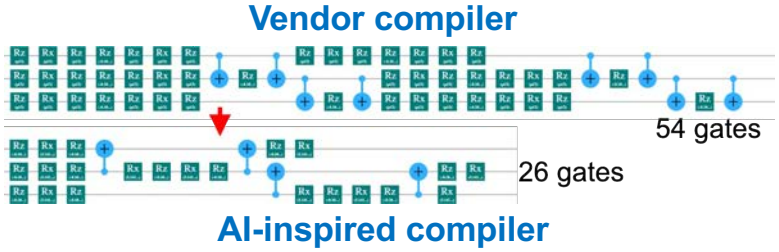
https://aiichironakano.github.io/phys516/08-1QuantumGates.pdf
https://aiichironakano.github.io/phys516/08-2TFIM.pdf

Quantum Dynamics on NISQ Computers

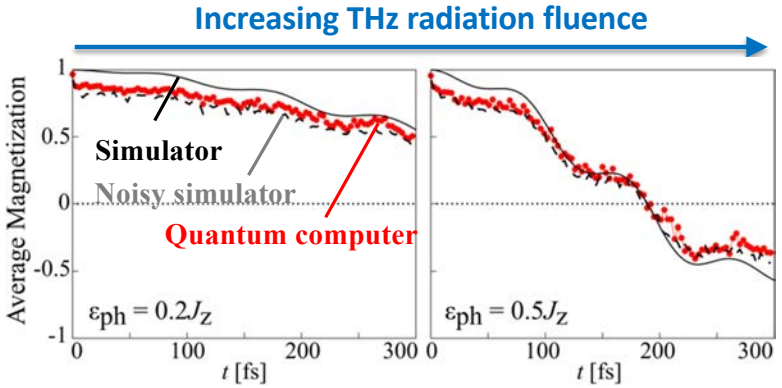
- **Quantum-dynamics simulations on NISQ computers show dynamic suppression of magnetization by THz radiation**

L. Bassman *et al.*,
Phys. Rev. B **101**, 184305 ('20)

- **AI-inspired quantum compiler reduced the circuit size by 30% to mitigate environmental noise**

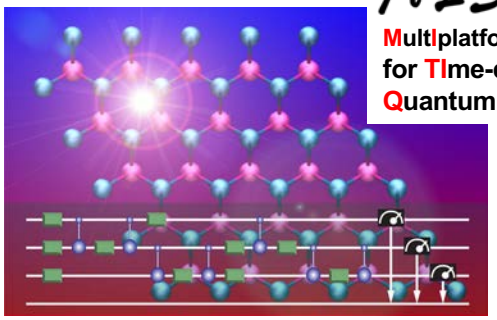


L. Bassman *et al.*,
Quantum Sci. Tech. **6**, 014007 ('21)



- **Full-stack, cross-platform software for quantum dynamics simulations on NISQ computers**

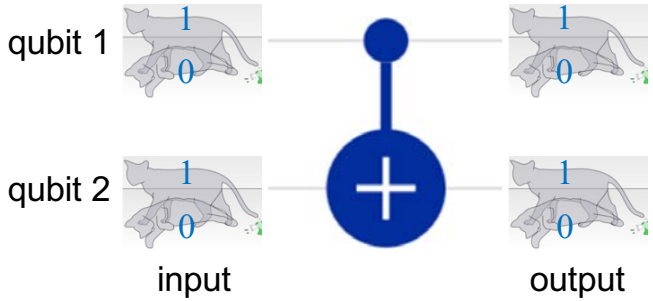
MISTIQS
Multi**I**platform **S**oftware for T**I**me-dependent **Q**uantum **S**imulation



C. Powers *et al.*, *SoftwareX* **14**, 100696 ('21)
https://github.com/USCCACS/MISTIQS

How Quantum Computing Works

- **Quantum parallelism: *Quantum* gate acting on *n*-qubit computes *all* 2^n *case works all at once*, thus achieving exponential speedup**
- **Example: Quantum CNOT (conditional not) gate**



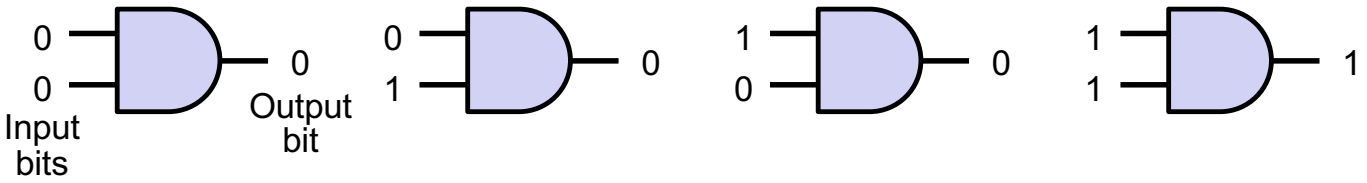
Mathematically, CNOT gate multiplies a 4×4 *matrix* to a $4(=2^2)$-element input *vector*

$$\text{input } \begin{matrix} 00 & 01 & 10 & 11 \end{matrix}$$

$$U_{\text{CNOT}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

output

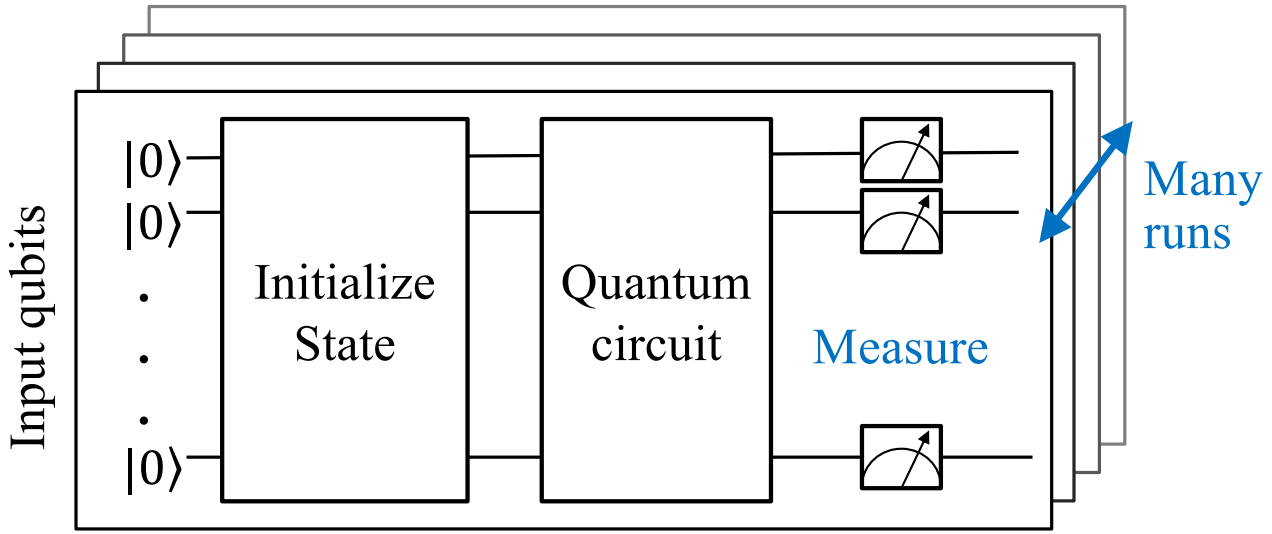
cf. Classical gate computes *only one case at a time*



Quantum Computing Runs & Measurements

- **While a qubit stored in a quantum register is a superposition of 0 & 1 states, once it is measured, the measured value is either 0 or 1, with the probability according to its quantum wave function**

/Users/aiichironakano/Documents/ANMB/ANMB-Viewgraphs/Movies/SchrodingerCat/schrodingers_cat_observation.html

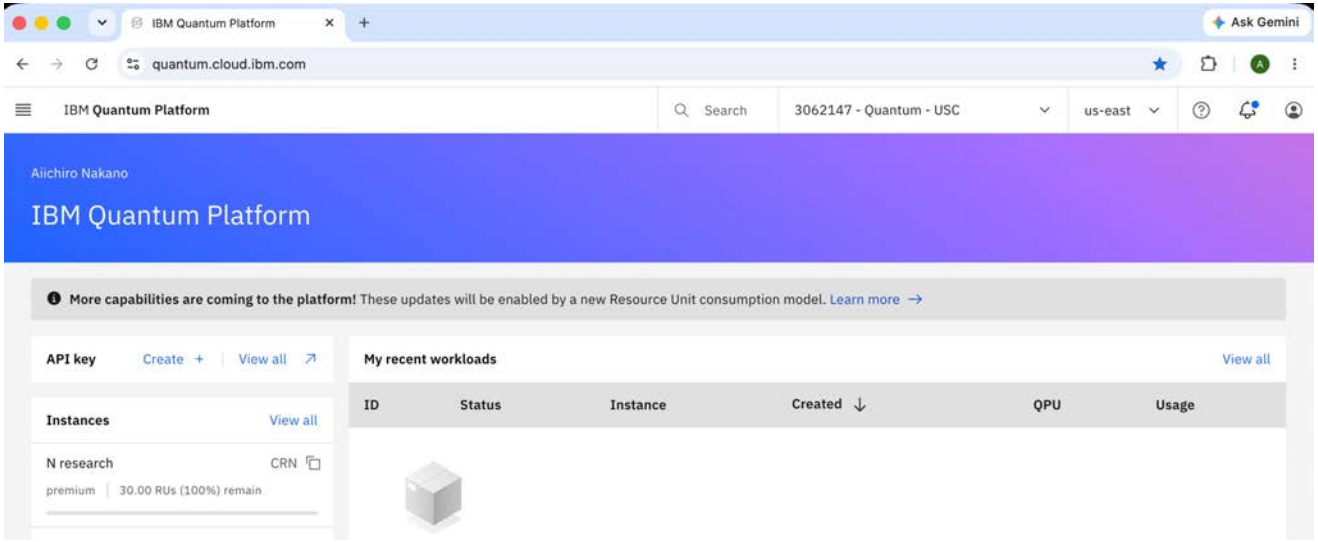


- **For each program, thousands of runs need be performed on a quantum computer to obtain statistics**

Next: Hands on at IBM Quantum

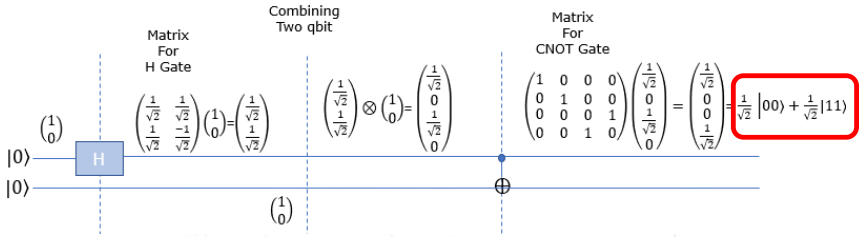
- **Quantum computing basics: Qubits and quantum gates**
- **Advanced: Quantum computation of transverse-field Ising model**

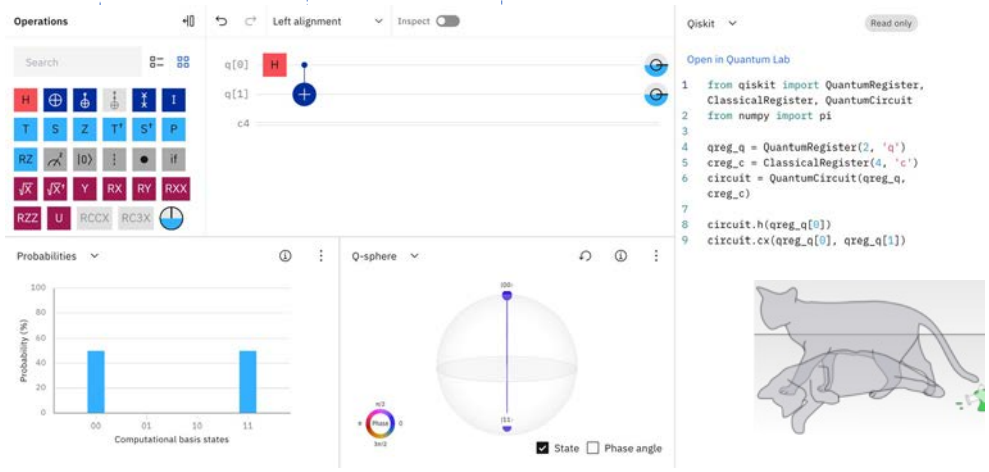
Do it yourself at https://quantum.ibm.com/

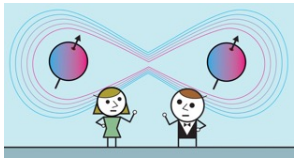


Do-It-Yourself Quantum Computing

- **Sign up & sign in to IBM Quantum cloud**
  https://quantum.cloud.ibm.com
- **Open Composer (GUI-based quantum programming platform)**
- **Build a quantum circuit to prepare an *entangled Bell's state*, which has played an essential role in quantum information science (2022 Nobel physics prize), using Hadamard & CNOT gates**







https://en.wikipedia.org/wiki/Bell_state

Hadamard gate transforms a pure qubit ($|0\rangle$ or $|1\rangle$) into a superposition of $|0\rangle$ & $|1\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

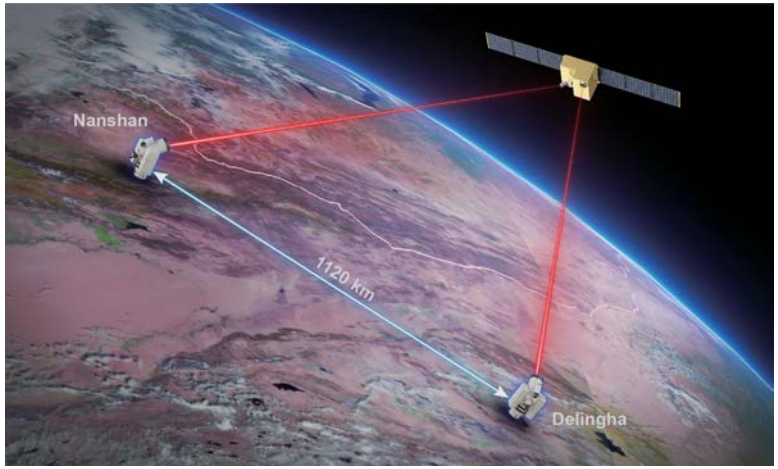
$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

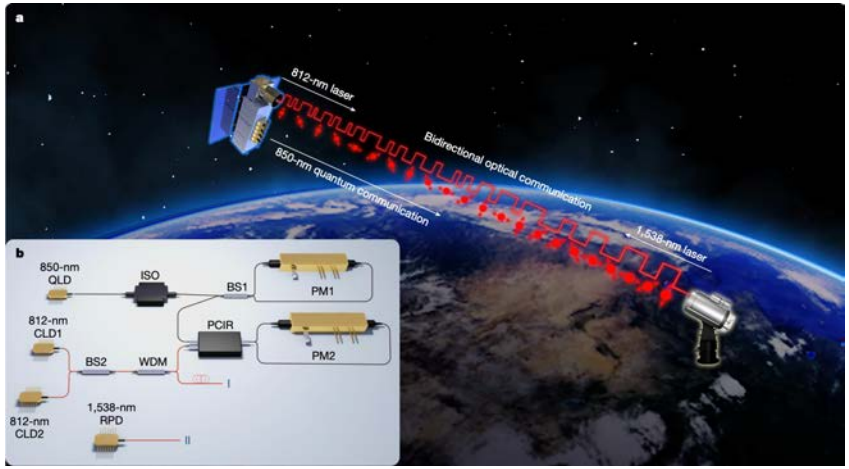
Entanglement: What's the Big Deal?

China's quantum satellite achieves 'spooky action' at record distance:
Result paves way for hack-proof quantum communications

Science Journal (June 15, '17)

Quantum entanglement—physics at its strangest—has moved out of this world and into space. In a study that shows China's growing mastery of both the quantum world and space science, a team of physicists reports that it sent eerily intertwined quantum particles from a satellite to ground stations ***separated by 1,200 kilometers***, smashing the previous world record. The result is a stepping-stone to ***ultrasecure communication networks and, eventually, a space-based quantum internet***. https://www.science.org/content/article/china-s-quantum-satellite-achieves-spooky-action-record-distance





Now quantum key distribution over *12,900 km* between China & South Africa *Nature* **640**, 47 ('25)

Quantum-Centric Supercomputing



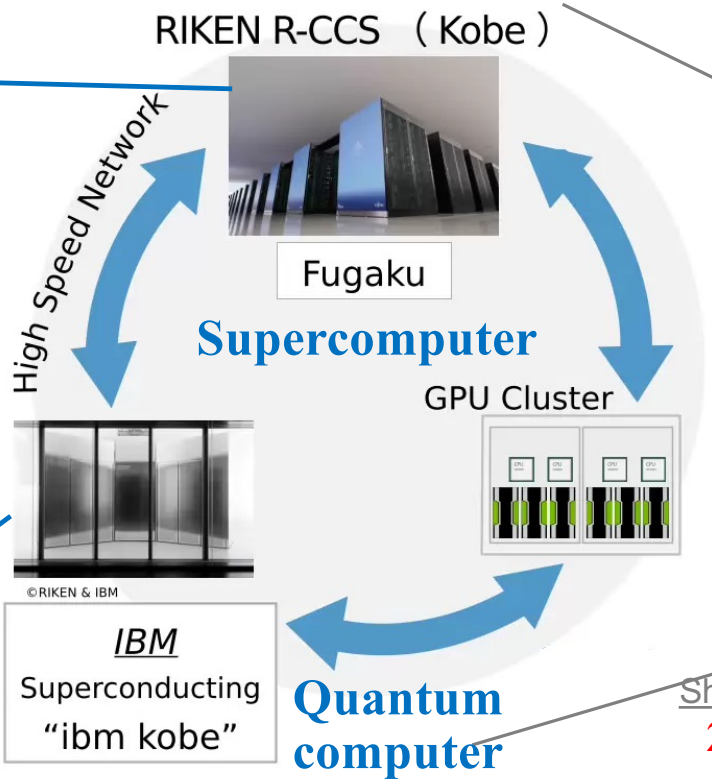
RunClassical()
C++/MPI

System call

RunQuantum()
Python/Qiskit



Hybrid quantum+supercompuring







State-of-the-art QCSC
Shirakawa *et al.*, *arXiv:* 2511.00224 (’25)
2026 Gordon Bell Prize finalist
Merz *et al.*, *arXiv*: 2605.01138v2

- **Many next-generation supercomputers will be tightly integrated with quantum computers as quantum processing unit (QPU), *e.g.*, FugakuNEXT at RIKEN, Japan & NERSC at Berkeley, U.S. in ~2030**

What Is Quantum Hard? Ask an Expert

- **Landmark computer science proof cascades through physics and math**
  Ji, Yuen *et al.*, *CACM* **64(11)**, 131 ('21)
  https://www.quantamagazine.org/landmark-computer-science-proof-cascades-through-physics-and-math-20200304/



4. MIP and quantum entanglement (this is the wild part)

MIP* = RE (major 2020 result)

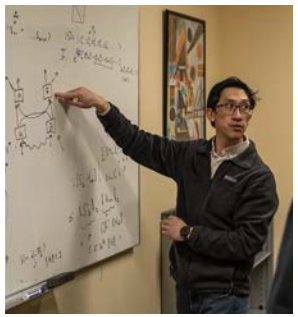
- **MIP*** = Multi-prover interactive proofs with entangled provers
- **RE** = Recursively enumerable problems (includes *undecidable* problems)

Translation:

> Allowing entangled provers makes the system so powerful it can verify problems that are **not even computable.**

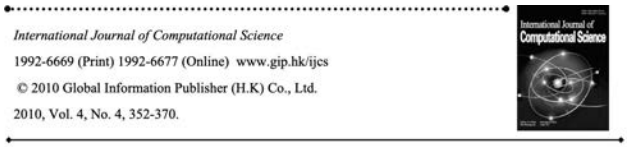
cf. NP: a solution can be verified by a classical Turing machine in polynomial time

Henry Yuen (*Columbia*)



- **A new complexity theory for the quantum age**
  https://www.quantamagazine.org/a-new-complexity-theory-for-the-quantum-age-20260217/

- **Prof. Henry Yuen's academic origin as a USC undergrad**
  https://arxiv.org/abs/1012.0900


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2010, Vol. 4, No. 4, 352-370.

DNA Sequencing via Quantum Mechanics and Machine Learning

Henry Yuen[1], Fuyuki Shimojo[1,2], Kevin J. Zhang[1,3], Ken-ichi Nomura[1], Rajiv K. Kalia[1], Aiichiro Nakano[1*], Priya Vashishta[1]

Where to Go from Here?

Try quantum-computing tutorials on IBM Quantum cloud
https://quantum.cloud.ibm.com/docs/en/tutorials

- **Install Python notebook (Jupyter) environment on your computer**
- **Get an IBM credential (it's free but to securely access actual quantum computers)**
- **Step-by-step video guide by Dr. Derek Wang (IBM)**
  https://www.youtube.com/playlist?list=PLOFEBzvs-VvrgHZt3exM_NNiNKtZlHvZi

Useful preparation

qubits quantum gates multi-qubits

- **Calculus & linear algebra (vector, matrix, tensor product)**
  —Watch a beautiful talk on physics, math & life by Prof. Ken Phillips (USC); *follow the math recipe!* (~1 hr. 15 min.)
  https://laist.com/events/the-story-collider-dec2024
- **Python programming**
  —Or ask ChatGPT Codex or Claude Code to write it for you

